

preprocessing of data

1. Dataset Description

The analysis uses a weekly city-level ecological dataset covering respiratory emergency department (ED) visits and air pollutant concentrations recorded in Milan, Italy. The dataset was compiled from administrative health records and air quality monitoring networks operated by ARPA Lombardia. Data were structured at the resolution of the ISO week, with one observation per week corresponding to the city as a whole.

The raw file contains 155 rows and 33 columns, including seven daily measurements per pollutant (Monday through Sunday), three health outcome counts, and two meteorological covariates. After preprocessing, the analysis-ready dataset comprises 155 weekly observations and 39 derived variables.

2. Study Period

The observation window spans from ISO Week 18 of 2023 (week starting 3 May 2023) to ISO Week 16 of 2026 (week ending 19 April 2026), covering approximately three calendar years. The total number of weekly observations is 155. The year 2023 is incomplete, beginning at W18, and should be flagged as such in any year-stratified analysis.

3. Weekly Aggregation of Exposure

Each pollutant was measured daily at city-level monitoring stations. For each week, the following summary statistics were computed from the seven daily values:

- Weekly mean (NO2_mean, PM25_mean, PM10_mean): primary exposure variable used in all models, computed as the arithmetic mean of available non-missing daily readings.
- Weekly maximum (NO2_max, PM25_max, PM10_max): secondary exposure metric.
- Weekly 75th percentile (NO2_p75, PM25_p75, PM10_p75): used as the exposure threshold in the Rc/WARM analysis (P3).
- Number of valid daily readings (NO2_n_days, PM25_n_days, PM10_n_days): quality indicator ranging from 0 to 7.

Weekly means were computed using available non-missing values (skipna=True), so weeks with fewer than 7 daily readings still contribute to the analysis, weighted by the number of valid days. This approach follows standard practice in air pollution epidemiology.

4. Missing-Data Handling

A systematic missingness audit was performed on all key variables. Results are summarised below.

Variable	N valid	N missing	% missing
NO2_mean	155	0	0.0%
PM25_mean	155	0	0.0%

PM10_mean	155	0	0.0%
temp_mean	155	0	0.0%
humidity_mean	155	0	0.0%
respiratory_disease	155	0	0.0%
ILI	155	0	0.0%
pneumonia	155	0	0.0%

No missing values were identified in any of the primary exposure, outcome, or meteorological variables. This confirms that the weekly aggregation procedure successfully produced a complete dataset suitable for time-series modelling.

5. Partial Week Handling

Week 52 of 2025 (W52/2025) is flagged as a partial exposure week (`w52_2025_flag = 1`) because daily pollutant records are available for only 5 out of 7 days, due to the calendar year boundary. The weekly mean was computed from the 5 available readings. This week is included in the main analysis but excluded in a dedicated sensitivity analysis to verify that results are not driven by the incomplete measurement window.

Week 8 of 2025 (W8/2025) is flagged as an outlier (`outlier_flag = 1`) because the respiratory disease count was 1,095 visits, representing a reduction of approximately 64% relative to the winter mean (3,071 visits/week). This anomaly is likely attributable to temporary under-reporting at the ED rather than a genuine epidemiological signal. This week is excluded from all primary analyses and from the winter-only sensitivity analysis.

6. Exposure Variables

Three air pollutants were considered as primary exposures, all expressed as weekly mean concentrations in $\mu\text{g}/\text{m}^3$:

- Nitrogen dioxide (NO_2): measured at urban traffic and background monitoring stations. Weekly mean: $33.1 \pm 12.0 \mu\text{g}/\text{m}^3$ (range: 12.8–66.1).
- Fine particulate matter ($\text{PM}_{2.5}$): particles with aerodynamic diameter $\leq 2.5 \mu\text{m}$. Weekly mean: $19.6 \pm 12.3 \mu\text{g}/\text{m}^3$ (range: 5.7–72.9).
- Coarse particulate matter (PM_{10}): particles with aerodynamic diameter $\leq 10 \mu\text{m}$. Weekly mean: $29.1 \pm 14.8 \mu\text{g}/\text{m}^3$ (range: 11.3–94.3).

A strong seasonal pattern was observed for all three pollutants, with markedly higher concentrations in winter (October–March) compared to summer (July–September): NO_2 winter mean 42.4 vs. summer 23.7 $\mu\text{g}/\text{m}^3$; $\text{PM}_{2.5}$ winter 27.4 vs. summer 12.0 $\mu\text{g}/\text{m}^3$; PM_{10} winter 37.7 vs. summer 21.1 $\mu\text{g}/\text{m}^3$.

$\text{PM}_{2.5}$ and PM_{10} exhibit very high collinearity (Spearman $\rho = 0.957$), and NO_2 is also highly correlated with both (NO_2 – $\text{PM}_{2.5}$: $\rho = 0.802$; NO_2 – PM_{10} : $\rho = 0.766$). For this reason, all statistical models use a single-pollutant specification; multi-pollutant models are not appropriate in this dataset.

7. Outcome Variables

Three weekly counts of ED visits were used as health outcomes:

- Respiratory disease (primary outcome): all respiratory-coded ED visits. Weekly mean: $2,648 \pm 693$ visits (range: 1,095–4,940). A marked winter excess was observed, with winter mean 3,071 vs. summer mean 1,956 visits/week.
- Influenza-like illness (ILI): ED visits coded as ILI or viral syndrome. Weekly mean: 932 ± 326 visits (range: 438–2,122). No structural zero-count weeks were identified.
- Pneumonia: ED visits with pneumonia diagnosis. Weekly mean: 373 ± 135 visits (range: 144–763).

All three outcomes are strictly positive across all 155 weeks, with no structural zeros. The distributions are right-skewed and consistent with a Poisson or quasi-Poisson generating process, supporting the use of log-linear models in the primary analysis.

8. Meteorological Covariates

Two meteorological variables are included as confounders in the regression models:

- Weekly mean temperature (temp_mean, °C): mean: $14.5 \pm 7.7^\circ\text{C}$ (range: 1.1–30.4°C). Temperature shows a strong inverse association with respiratory disease (Spearman $\rho = -0.809$), reflecting the expected winter-peaking pattern of respiratory morbidity.
- Weekly mean relative humidity (humidity_mean, %): mean: $76.9 \pm 11.0\%$ (range: 32.0–94.9%).

Important modelling note: temperature is included as a within-season confounder, not as a proxy for seasonality. In the DLNM (P2), temperature is modelled using a natural cubic spline $\text{ns}(\text{temp_mean}, \text{df}=3)$ to capture non-linear residual within-season variation, while the periodic seasonal cycle is controlled by Fourier harmonics ($\sin 52$, $\cos 52$, $\sin 26$, $\cos 26$). Using temperature as the sole seasonal control leads to over-adjustment and attenuates the pollutant effect estimates.

9. Descriptive Statistics

Table 2 presents the full descriptive summary for all primary variables across the 155-week study period (outlier W8/2025 included for completeness; analyses exclude this week).

Variable	Mean	SD	Min	P25	Median	P75	Max
NO ₂ (µg/m ³)	33.1	12.0	12.8	22.9	31.2	42.0	66.1
PM _{2.5} (µg/m ³)	19.6	12.3	5.7	11.2	14.7	24.6	72.9
PM ₁₀ (µg/m ³)	29.1	14.8	11.3	19.4	24.4	34.5	94.3
Temperature (°C)	14.5	7.7	1.1	7.9	14.6	20.9	30.4

Humidity (%)	76.9	11.0	32.0	70.0	78.6	85.5	94.9
Respiratory disease (visits/wk)	2648	693	1095	2289	2656	2886	4940
ILI (visits/wk)	932	326	438	722	882	985	2122
Pneumonia (visits/wk)	373	135	144	276	358	429	763

Table 2. Descriptive statistics for primary variables. N = 155 weekly observations (W18/2023–W16/2026). SD = standard deviation; P25/P75 = interquartile range.

10. Main EDA Findings

10.1 Seasonal patterns

All three pollutants and all three outcomes display a pronounced seasonal cycle. Pollutant concentrations peak in winter, driven by reduced atmospheric mixing, increased residential heating, and lower photochemical degradation of NO₂. Respiratory ED visits are also highest in winter, consistent with the combined effects of cold temperature, increased indoor crowding, and higher pollutant exposure.

The Spearman correlation between temperature and respiratory disease is $\rho = -0.809$ ($p < 0.001$), confirming that temperature is a dominant seasonal signal. This must be accounted for separately from the pollutant effect in the regression models.

10.2 Exposure–outcome associations

Positive Spearman correlations were observed between all pollutants and respiratory disease at the unadjusted level: NO₂ ($\rho = 0.649$), PM_{2.5} ($\rho = 0.503$), PM₁₀ ($\rho = 0.468$). These raw associations are partially driven by the shared seasonal pattern with outcomes and are not estimates of causal effects; adjustment for seasonality and temperature is essential before interpreting pollutant coefficients.

10.3 Collinearity

The three pollutants are highly inter-correlated. The PM_{2.5}–PM₁₀ pair exhibits very high collinearity ($\rho = 0.957$), which precludes their simultaneous inclusion in any regression model. NO₂ is also highly correlated with both particulate measures (NO₂–PM_{2.5}: $\rho = 0.802$; NO₂–PM₁₀: $\rho = 0.766$). All models must therefore adopt a single-pollutant specification.

10.4 Outlier W8/2025

A visual inspection of the respiratory disease time series identified one anomalous observation in Week 8 of 2025, with a count of 1,095 visits (z-score: -2.2 relative to the winter distribution). Pollutant concentrations in the same week were unremarkable, suggesting that the anomaly reflects an administrative recording gap rather than a genuine health event. This week is excluded from all primary analyses.

11. Key Data Limitations

- Ecological design: all analyses are performed at city level. Exposure and health outcomes are city-wide aggregates; no individual-level or district-level inference is possible. Ecological bias may affect the magnitude of estimated associations.
- Single monitoring network: pollutant concentrations represent a city-average derived from available ARPA Lombardia stations. Spatial heterogeneity in exposure within Milan cannot be captured at this resolution.
- Incomplete first year: the dataset begins at W18/2023, providing only approximately 35 weeks of observations for 2023. Year-stratified analyses should flag 2023 as an incomplete year.
- Partial final week: W52/2025 contains only 5 of 7 daily pollutant readings. Although included in the main analysis, results should be verified through the sensitivity analysis excluding this week.
- Fixed population denominator: outcome counts are not expressed as rates because Milan's population is assumed constant across the study period. Minor year-to-year population changes are not accounted for.
- No individual-level covariates: age, sex, and socioeconomic status cannot be controlled for in this ecological design. Residual confounding by population composition changes is possible but assumed minimal over the three-year window.